RIPHAH INTERNATIONAL UNIVERSITY ISLAMABAD

GULBERG GREEN CAMPUS

The Closet Fashion Assistant

(AI-Based Smart Outfit Recommendation System)

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Project Proposal

Project Title:

The Closet Fashion Assistant: An Intelligent AI-Based Outfit Recommendation Agent

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Description:

In today's busy lifestyle, individuals spend a significant amount of time deciding what to wear. Many people face difficulty in selecting outfits that match weather conditions, occasion or event, seasonal clothing requirements, personal style preferences, and clean/dirty clothing status. Existing fashion platforms lack **personalized wardrobe integration**, **real-time weather awareness**, and **intelligent outfit combination logic**.

The Closet Fashion Assistant bridges this gap by creating an AI-powered personal stylist that lives in your pocket. This system will:

- Automatically detect **real-time weather** of user's saved location using API integration
- Maintain **separate seasonal wardrobes** (summer/winter) with uploaded clothing images
- Recommend outfits using **machine learning (KNN)** and _A search algorithm_*
- Track **clean/dirty clothing status** to avoid suggesting unavailable items
- Display **combined outfit preview images** (shirt + pants + shoes) for better visualization
- Learn from **wear history** to prevent outfit repetition

Target audience : Students, working professionals, fashion-conscious individuals, and anyone seeking time-efficient dressing decisions.

Major Features:

FE-1: One-Time Location Setup – User enters location once. System automatically detects weather of that location using weather API.

FE-2: Seasonal Wardrobe Management – Separate wardrobe sections for summer and winter clothes. User uploads clothes according to current season.

FE-3: AI-Based Outfit Recommendation – System uses Machine Learning (KNN) and A* Search algorithm with utility scoring function to select best outfit.

FE-4: Automatic Weather Detection – System automatically fetches real-time weather of user's saved location.

FE-5: Occasion-Based Filtering – User selects event type (casual, formal, interview, wedding, party, etc.)

FE-6: Clean/Dirty Tracking System – Selected clothes are automatically marked dirty. Laundry Mode allows user to mark clothes clean again.

FE-7: Combined Outfit Image Preview – System displays a single combined image showing shirt, pants, and shoes together for visualization.

FE-8: Accept / Try Another Option – User can accept recommended outfit or request an alternative recommendation.

FE-9: Wear History Tracking – System maintains history of worn outfits and avoids frequent repetition.

FE-10: Performance Evaluation Metrics – System evaluates ML model accuracy and algorithm efficiency using confusion matrix and accuracy scores.

System Studied:

SYS-1: Pinterest Fashion Recommendation – https://www.pinterest.com
SYS-2: Google Outfit Suggestions – https://www.google.com/shopping
SYS-3: Amazon Fashion AI – https://www.amazon.com/fashion
SYS-4: Zara Virtual Styling – https://www.zara.com
SYS-5: Myntra Style Recommendations – https://www.myntra.com

Table:

Features	Pinterest	Amazon	Zara	Myntra	Closet Assistant
Weather Detection	N	N	N	N	**Y**
Seasonal Wardrobe	N	N	N	N	**Y**
AI Recommendation	Y	Y	Y	Y	**Y**
Clean/Dirty Tracking	N	N	N	N	**Y**
Image Preview	Y	Y	Y	Y	**Y**
A* Search Algorithm	N	N	N	N	**Y**
User Location Setup	N	N	N	N	**Y**
Occasion-Based Filtering	Y	Y	Y	Y	**Y**
Wear History Tracking	N	N	N	N	**Y**

Keys:

Y = Feature Present
N = Feature Absent